Review of smart systems in aquaculture and aquaponics

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***Abstract*—In this paper, we present a set of works between 2019-2025 in the field of aquaculture and smart aquaculture systems, which witnessed remarkable developments by integrating modern technologies such as artificial intelligence, Internet of Things and cloud computing. The works were based on a set of standards to monitor and automatically control environmental changes in the system and maintain its stability, and the works presented a set of technologies in the field of tracking the growth of crops including fish and plants and enhancing the efficiency of resource management to reduce waste. In addition to works that included proactive prediction of environmental variables, decision support systems for failure analysis, and automated feeder systems, with the aim of developing smart systems that address new challenges and solve previous shortcomings observed in the field.**

Keywords— aquaponics, Aquaculture, Internet of Things, Artificial Intelligence, Smart Systems

I. Introduction

With the spread of wars and global crises, nearly one in ten people were exposed to severe levels of food insecurity in 2019. The latest estimates indicate that over 8.8% of the world's population suffers from hunger [1].And with population growth and the agricultural labour power shortage, it has become important to provide solutions that achieve an economic alternative to deal with these crises. Aquaponics systems are one of the solutions and the most important food production systems at the present time, as they include the economy of resources in a system consisting of two units in [2], an aquaculture system and an irrigation and agricultural fertilisation system within a closed system that achieves environmental balance, as the wastewater of the aquaculture system is rich in nutrients useful for the plant [1], but manual labour and reliance on traditional methods are expensive and not protected from human errors. For example, aquaculture water quality parameters are typically assessed through laboratory analyses, which are costly and time-consuming [3]. And therefore this field is witnessing a major shift in the methods used, and with the rapid technological development. IoT and AI technologies can address these challenges by automating monitoring and control, analysing sensor data, and detecting patterns that may not be easily recognised by humans [4]. In this context, this research reviews an analysis of the latest innovations during the period 2019-2025 that introduced modern technologies supporting aquaponics systems and aquaculture, assesses the challenges, and proposes new methods.

II. Comparative analysis of previous works

A. Monitoring and control using the Internet of Things

Internet of Things (IoT) technologies are among the most important monitoring technologies in smart systems today in the fields of aquaculture, irrigation, and aquaponics systems. Several modern technologies can be observed in Table 1 in the field of smart monitoring, including sensors, cloud platform support, and microcontrollers such as Arduino and NodeMCU. The aim is to identify and address problems as quickly as possible and reduce human intervention. In addition, integrating the work of [5]the WeMos D1 Wi-Fi Uno (ESP8266) microcontroller for data processing and transmission via Wi-Fi. The MQTT protocol was employed in the study by [6],with the aim of ensuring secure communication between devices and the cloud to enable the integration of artificial intelligence and to automatically improve system standards, while platforms like Ubidots and Thingspeak were used in the studies of [7] and [8] for data collection and real-time display. In [6], solar energy was employed, which was beneficial for the environment and the energy economy, especially in farms. Additionally, [9], developed an SMS-based alert system to notify users of any system malfunction.

TABLE. 1 TECHNOLOGIES AND TOOLS FOR REMOTE MONITORING AND CONTROL IN AQUACULTURE AND AQUAPONICS SYSTEMS

Study	Algorithms and Processing Techniques	Control Devices
[5]	Remote control using the Blynk application	WeMos D1 Wi-Fi Uno (ESP8266)
[6]	MQTT protocol for data analysis and security	NodeMCU with solar power systems
[7]	Data visualization using the Ubidots platform	NodeMCU
[8]	Data collection and analysis via Thingspeak	Arduino Uno
[9]	SMS alert system (GSM)	Arduino Mega

In the presented monitoring and control systems, sensors are important elements. In previous works in Table 2, it was

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observed that most studies used temperature and humidity sensors and pH sensors to ensure an ideal environment for plants and fish, and as a key tool for monitoring water quality. Water level sensors were also used to avoid problems of water shortage or excess. Works such as [6] distinguished themselves by using CO_2 and ammonia sensors to monitor the quality of nutrients in the water.

TABLE. 2 COMPARISON OF SENSORS USED IN MONITORING AND CONTROL FOR PREVIOUS WORKS IN AQUACULTURE AND AQUAPONICS SYSTEMS

Study	Temperature & Humidity Sensor	pH Sensor	Water Level Sensor	Soil Moisture Sensor	Sensors Ammonia	Sensors CO_2
[5]	✓	✗	✓	✓	✗	✗
[6]	✓	✓	✓	✗	✓	✓
[7]	✓	✓	✓	✗	✗	✗
[8]	✓	✓	✓	✗	✗	✗
[9]	✓	✓	✓	✗	✗	✗

Figure 1 illustrates the architecture of the Internet of Things (IoT) system used for monitoring in the aquaponics systems of [7] work. It consists of three layers: first, the physical layer includes sensors that capture data; then, the gateway layer consists of a NodeMCU device as an intermediary for transmitting data via the MQTT (Message Queuing Telemetry Transport) protocol. The data is transmitted via MQTT/WebSocket and directed to the MQTT Broker, then to the IoT Cloud for storage and analysis. Then comes the application layer, which includes a user interface that allows data to be displayed in various forms.

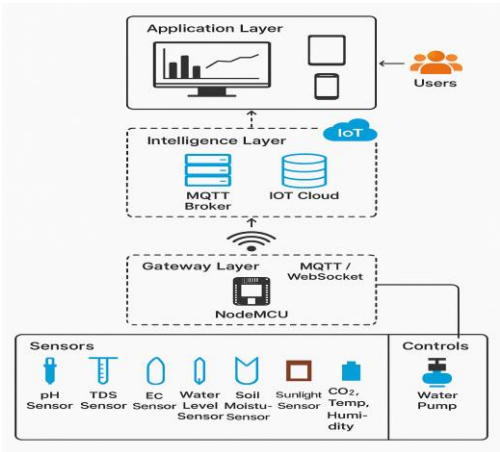

Fig. 1. The surveillance system and the Internet of Things in operation [6].

These initiatives contributed to the results shown in Table 3. Close monitoring, as in [7]; adding a control interface in [5]; automated control in [8] based on scientific data; supporting communication security in [6]; and building warning systems in [9]. However, the variation in the standards used reveals gaps in the monitoring process, in addition to the absence of a monitoring system for plant and aquatic organism health, and the fact that human remote control is prone to error and the automatic control provided in [8] relies on scientific values that lack intelligence and cannot achieve a balance between different criteria. These systems lack monitoring of the complex interactions between fish, plants, bacteria, and the environment, and monitoring of the balance between environments.

TABLE. 3 RESULTS OF REMOTE MONITORING AND CONTROL IN AQUACULTURE AND AQUAPONICS SYSTEMS

Study	Key Results
[5]	Reduced labor dependence with enhanced plant and fish productivity.
[6]	The architecture is designed to be scalable, secure, and solar-powered. Relies on the secure MQTT protocol for communication.
[7]	High sensor accuracy, enhanced lettuce and fish growth despite heat challenges.
[8]	Strong relationship between time and plant height and fish weight ($R^2 > 0.96$).
[9]	Stable pH range (6.5–8) with consistent plant growth improvement.

B. Automated and intelligent control of Aquaponics systems

To enhance the efficiency of aquaculture and hydroponics systems, several studies have proposed initiatives to build a smart system that operates with automatic control using the technologies described in Table 4. Systems have been developed to improve water quality and preserve the environment of living organisms using technologies that mostly incorporate fuzzy logic due to its ability to adapt and deal with inaccurate and ambiguous data in the environment, and traditional techniques that rely on experts and precise equations, such as PID (Proportional-Integral-Derivative) control, contributed to achieving good results in maintaining the environment of the aquaponics system. Additionally, the work of [10] and [11] included a biofilter for the growth of specialized nitrifying bacteria in the nitrification process, which is the conversion of ammonia (NH_3) or ammonium (NH_4) to nitrate (NO_3-) to improve the nutrients in the water. The most important task of the system was to provide a medium for the life of these bacteria and to maintain the quality of the nutrient solution to contribute to the health of the plant. [12] work presented ideas to improve water conditions using Genetic Algorithms (GA) to predict the best water conditions, as well as Random Forest modeling to respond to plant growth based on water criteria. These works have enhanced their systems to support automated control through integration with the Internet of Things and cloud platforms to process and analyze data in the best possible way. MATLAB Simulink was used in [10] and [11] for the simulation and application of mathematical algorithms, AWS Cloud for remote monitoring, Raspberry Pi as the main processing unit for local analysis, and controllers such as Arduino and ESP8266 for wireless communications. On the other hand, the results in Table 5 show that these intelligent systems have significantly improved system performance. These systems have also contributed to the effective control of resources, productivity, and the health of living organisms.

TABLE. 4 SMART CONTROL TECHNIQUES AND TOOLS IN PREVIOUS WORKS IN AQUACULTURE AND AQUAPONICS SYSTEMS

Study	Algorithms	Processing Technology	Control Mechanism
[10]	Fuzzy Logic, (Membership Functions, 9 Fuzzy Logic Rules)	Arduino, Raspberry Pi, MATLAB Simulink	Peltier Elements Solenoid Valve
[11]	PID Control Algorithm	ESP8266, MATLAB Simulink	DC water pump Water heater Aerator Automatic fish feeder Solenoid valves LED plant lighting
[12]	Random Forest, Genetic Algorithm	Arduino Uno microcontroller, PIC microprocessor	Nema 17 stepper motors, motor modules, Chemical Dosing Systems
[13]	Fuzzy Logic, (Membership Functions, 9 Fuzzy Logic Rules)	Arduino, eFLL (Embedded Fuzzy Logic Library)	Solenoid Valve
[14]	Fuzzy Logic (Membership Functions,17 Fuzzy Logic Rules)	ESP32, Arduino Mega2560 Pro Mini	Water Pump, Proportional Valve, Peristaltic DC Pump Heater
[15]	Fuzzy Logic, Membership Functions	Cloud-based data transfer (AWS), Node MCU ESP32	Control of actuators DC water pump L298 motor(aerators)
[16]	Fuzzy Control System	Arduino and Raspberry Pi	Air pumps and automatic feeding system
[17]	Fuzzy Logic (Decision-making), IoT	Arduino ESP8266, Raspberry Pi Oracle APEX on AWS	Heater, Aerator, Peristaltic Pump L298N Motor Driver, AC Dimmer

TABLE. 5 RESULTS OF SMART MONITORING AND CONTROL IN AQUACULTURE AND AQUAPONICS SYSTEMS

Study	Results
[10]	Stabilized pH (7-8) and temperature (25-30°C) to optimize the environment for nitrifying bacteria.
[11]	Water quality stabilization, resource efficiency improvement, and error reduction using PID.
[12]	Stable water conditions, enhanced productivity.
[13]	Maintained optimal pH (6.0-6.5) and EC (0.8-1.2 mS/cm); enhanced plant growth.
[14]	Restored water quality (temperature: 32°C, pH: 6-8) efficiently and quickly; improved fish productivity.
[15]	Achieved 98% accuracy; remote monitoring via mobile app; enhanced aquatic organisms' environment. And it demonstrated the superiority of fuzzy logic over ANFIS and ANN algorithms.
[16]	Increased shrimp survival rate by 33.3% compared to traditional methods.
[17]	Increased shrimp survival by 6%, improved growth rate by 28%.

In Figure 2, the aquaponics system from [11] Fish waste is converted by a mechanical and biological filter into nitrates that nourish the plants, while the plants purify the water to be recycled for the fish. And water and air pumps to ensure continuous flow and aeration, in addition to a food dispenser for feeding the fish, and sensors used for monitoring. The system is intelligently controlled through a PID algorithm, which adjusts the system when there is a deviation from the ideal values. LED lighting enhances plant growth.

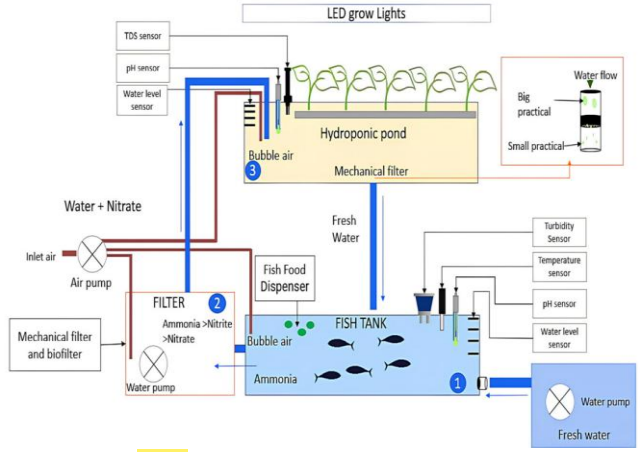

Fig. 2. Aquaponics model for work [11]

Figure 3 illustrates the decision-making model by [15], based on fuzzy logic used in some of the mentioned studies. The dataset captured by the sensors undergoes a fuzzification phase by converting the data into fuzzy values, followed by fuzzy inference based on rules and fuzzy functions, and then defuzzification to convert the results into clear decisions. The outputs include turning on/off the aerator or the fresh water pump.

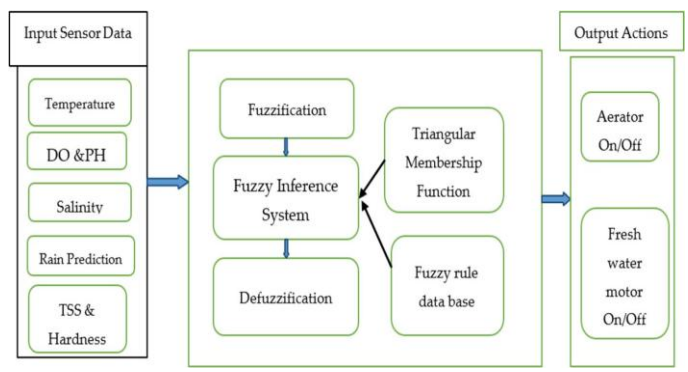

Fig. 3. The mechanism of automatic control using fuzzy logic in the operation of [15].

Despite what has been achieved in the context of automatic and intelligent control, we note that most of these achievements rely heavily on fuzzy logic and PID, which are similar in principle. It is true that they are effective in dealing with ambiguous data or maintaining and ensuring speed because they are lighter than artificial intelligence models, but they rely on ready-made knowledge such as functions, fuzzy rules, and predictive modeling based on a specific type of object. In addition, we note in Table 6 that most studies have focused on pH and temperature while neglecting DO (dissolved oxygen), which is important for fish health, and the absence of biological indicators, neglecting the importance of nitrate, ammonia, and even water hardness (Ca^{2+} Mg^{2+}), and neglecting climatic changes and external environmental conditions, given that aquaculture and aquaponics systems are often in an open environment, as well as relying on cloud computing, which is insufficient because smart control depends on response speed,

which necessitates the importance of local processing that is closer. We also did not observe systems to avoid expected failures.

TABLE. 6 COMPARISON OF SENSORS USED IN SMART CONTROL OF PREVIOUS WORKS IN AQUACULTURE AND AQUAPONICS SYSTEMS

Study	pH Sensor	Temperature Sensor	Turbidity Sensor	EC Sensor	TDS Sensor	Water Level Sensor	DO Sensor	Salinity Sensor	Water Hardness Ca^{2+} Mg^{2+}	Nitrite Sensor
[10]	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗
[11]	✓	✓	✓	✗	✓	✓	✗	✗	✗	✗
[12]	✓	✓	✗	✗	✗	✗	✓	✓	✗	✓
[13]	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗
[14]	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗
[15]	✓	✓	✓	✗	✗	✗	✓	✓	✗	✗
[16]	✓	✓	✗	✗	✗	✓	✗	✗	✓	✗
[17]	✓	✓	✗	✗	✗	✗	✓	✗	✗	✗

C. *Decision support in aquaculture and aquaponics systems*

Decision support and optimization techniques are an important axis in intelligent systems as they use a set of technologies that include data analysis and transformation into models that help in making informed decisions. Table 7 represents part of these technologies in the field of aquaponics and aquaculture. For example, [18] focused on developing a system using artificial intelligence and the Internet of Things to support positive decision-making in aquaculture farms by collecting data on water quality and nutrition levels. A deep learning model (DNN) was developed using Bayesian optimization to tune parameters, predict fish growth, and control feeding In contrast, [19] presented an intelligent cultivation system based on analyzing farmers' needs using QFD, which facilitated farm management and achieved user satisfaction [20] developed a decision support system based on digital twins to simulate aquaponic systems in real-time. The system aims to coordinate distributed urban agriculture systems to improve production and reduce resource waste. Algorithms such as linear regression, support vector machines, and decision trees were applied to predict the growth rates of plants and fish. The work relied on the use of the sensors represented in Table 9 for monitoring and data collection. The contribution of the works can be observed in Table 8. Figure 4 shows the design of [20] aquaponics system, which includes a set of elements that intervene in making smart decisions. Among these are the inputs, represented by the sensors used for monitoring, and the outputs, which include control devices for regulating pH levels and solutions to increase or decrease acidity, as well as an air pump. The design also features biological components such as a biological filter and nitrification bacteria to convert ammonia to nitrite (Ammonia > Nitrite > Nitrate) for irrigating plants.

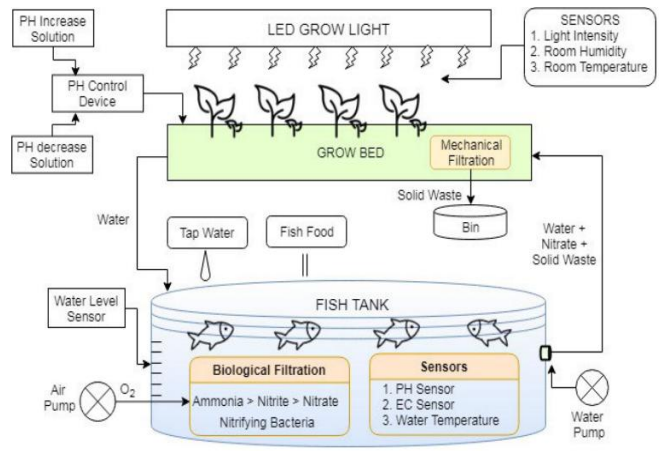

Fig. 4. Aquaponics model for work [20]

TABLE. 7 DECISION SUPPORT TECHNIQUES AND TOOLS IN AQUACULTURE AND AQUAPONICS SYSTEMS FOR PREVIOUS WORKS

Study	Algorithms	Processing Units	Control
[18]	Deep Neural Networks (DNN) with Bayesian Optimization	Arduino Mega2560 with Wi-Fi module	Water pump, heater, agitator, smart feeder
[19]	QFD (Quality Function Deployment), IoT	IoT Sensors, Cloud Computing	Automatic feeders, real-time monitoring systems
[20]	Digital Twin Simulation	IoT and cloud computing platforms	Water pumps, lighting system (LEDs), flow control valves

TABLE. 8 THE CONTRIBUTION OF WORKS TO SUPPORTING DECISION-MAKING IN AQUACULTURE AND AQUAPONICS SYSTEMS

Study	Results
[18]	Achieved high prediction accuracy for fish growth (R^2 = 0.94), reduced feed waste, improved water quality monitoring.
[19]	Improved water quality, increased production efficiency, user satisfaction above 4/5.
[20]	Enhanced urban aquaponics optimization, reduced resource waste, and real-time performance predictions.

TABLE. 9 COMPARISON OF THE CRITERIA USED IN DECISION-MAKING FOR PREVIOUS PROJECTS IN AQUACULTURE AND AQUAPONICS SYSTEMS.

Study	Dissolved Oxygen Sensor	Temperature Sensor	Turbidity Sensor	pH Sensor	Proximity Switch	Soil Moisture Sensor	Light Sensor	Water Flow Sensor	Total Dissolved Solids (TDS) Sensor	Salinity	Ultrasonic sensor (HC-SR04)
[18]	✓	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗
[19]	✗	✓	✗	✓	✗	✗	✗	✗	✗	✓	✓
[20]	✗	✓	✗	✓	✗	✗	✓	✓	✓	✗	✗

The systems in these works are generally based on physical and chemical indicators, but they are not linked to biological indicators (fish health, feeding behaviour, bacterial activity,

disease resistance) and decision-making criteria. Decision-making criteria must be comprehensive, including weather criteria, in order to build more accurate prediction models.In addition, no automatic intelligent control algorithms were mentioned.

D. *Techniques for improving aquaponics systems and smart aquaculture*

Improving aquaponics and aquaculture systems using modern technologies aims to maintain system quality, enhance productivity, and conserve resources. Some works have presented solutions and techniques that can be observed in Table 10. [21] presented the DRAPS system, which achieved exceptional efficiency in resource consumption and high productivity through a dual water recirculation system that separates fish tanks and plant cultivation, supported by artificial intelligence technologies. On the other hand, [22] provided solutions for sustainability by integrating a Recirculating Aquaculture System (RAS) and a Zero Water Discharge (ZWD) system, relying on the measurement of Temperature, Dissolved Oxygen, and Ammonia, which resulted in reduced water discharge and energy consumption while maintaining high survival rates for shrimp. On the other hand,[23]presented ideas for improving nutrient concentration using a machine learning algorithm to reduce harvest time and promote lettuce growth . They used 21 variables in their study, demonstrating the strength of the results. Table 11 contains the results of this work, as they reflect good solutions in resource efficiency and production improvement, with each focusing on distinctive aspects of smart and sustainable aquaponics systems.

TABLE. 10 TECHNIQUES AND TOOLS FOR IMPROVING AND DEVELOPING AQUACULTURE AND AQUAPONICS SYSTEMS FOR PREVIOUS PROJECTS

Study	Algorithms	Processing techniques
[21]	Long Short-Term Memory (LSTM) Bayesian Network (BN) Convolutional Neural Network (CNN) Zero-Positive Learning	IoT-Cloud system for data collection and sharing, optimization unit, digital diagnostics unit (Digital Sentinel).
[22]	Statistical analysis of shrimp growth and feed conversion ratio (FCR)	Multi-cycle hydraulic system (5 water cycles: sedimentation, protein skimmer, activated carbon, mechanical and biological filters). Central monitoring system with automatic cleaning of pipes and sensors.
[23]	Linear Support Vector Machine (SVM) Linear Discriminant Analysis (LDA) Decision Trees (CART) K-Nearest Neighbors (KNN)	Dimensionality reduction techniques (PCA, Feature Selection using XGBoost, ExtraTreesClassifier). Data collected from 3 commercial farms (Texas).

TABLE. 11 CONTRIBUTION TO THE IMPROVEMENT AND DEVELOPMENT OF AQUACULTURE AND AQUAPONICS SYSTEMS.

Study	Results
[21]	20% productivity improvement, 97% reduction in water consumption.
[22]	27% energy saving, 90% shrimp survival rate.
[23]	Increased crop size and quality, reduced harvest time to 4 weeks.

E. *Predictive and intelligent analysis models*

Prediction systems are used to analyze environmental data using techniques such as machine learning and deep learning. In the field of ecoponics, several works have made various contributions, such as the work of [24], which included classifying the growth stages of plants like lettuce by reducing the number of features and testing deep algorithms. In addition to the work of [25], which relied on criteria such as pH and temperature. To predict the types of fish capable of living in a specific environment. There are works that have focused on predicting water quality, such as [3], which aimed to predict water quality and detect "harmful algal blooms." To prevent its spread and thus improve salmon farms. And also the work of Rasheed [26] ,which supported the idea of predicting water quality parameters in real-time by reducing computation time. The algorithms and data used can be observed in Tables 12 and 13. The prediction techniques included all the basic elements in aquaponics systems, such as fish, water quality, and plants, but there was no study that combined the elements to achieve balance and stability for each element. We also noted a lack of biodiversity. The data ranged from only 191 samples, or 300 images, and the time frame of the study was short, which is somewhat insufficient for training deep learning algorithms. Nevertheless, the results provided acceptable accuracy.

TABLE. 12 COMPARISONS OF FORECASTING TECHNIQUES IN PREVIOUS WORKS ON AQUACULTURE AND AQUAPONICS SYSTEMS

Study	Dataset Size	Data
[3]	10 weeks	Chlorophyll-a (470 nm, 530 nm), turbidity
[24]	300 images	Morphological features
[25]	191 samples	pH (6.5–9), temperature, turbidity
[26]	Multiple datasets (ADAK, MAC)	Salinity, pH, dissolved oxygen (DO), temperature

TABLE. 13 COMPARISONS OF THE DATA USED IN FORECASTING PREVIOUS WORKS

Study	Algorithms	Performance Results (Accuracy)	Key Metrics
[3]	EEMD-MLR-LSTM	MAE: 0.0375, MAPE: 0.0072	MAE, RMSE, MAPE
[24]	LR, KNN, L-SVM	KNN: 91.67%	Accuracy, Cross-validation
[25]	Random Forest, J48, KNN	RF: 88.48%	Accuracy, Kappa
[26]	CNN-LSTM, CNN-GRU	CNN-LSTM outperformed all models	MAE, MSE, RMSE

F. *Intelligent and automated feeding systems*

Intelligent feeding systems are a key element of hydroponics and a vital axis for organism survival. They also significantly reduce resource economy and human error. [27]) created a smart feeding system that employs load cell sensors to measure exact weight and the ARIMA model to forecast future feeding requirements in order to increase feeding accuracy in aquaponics systems. In [28] introduced an automated feeding system based on fuzzy logic, taking into account environmental factors such as temperature and pH, using ultrasound to monitor water and feeding schedules. [29] study integrated computer vision with YOLOv5 and. fuzzy neural networks (FNN) to effectively analyze the feeding behavior of fish in aquaculture systems. The techniques used can be referenced in Tables 14.

TABLE. 14 COMPARISON BETWEEN SMART SYSTEMS FOR FEEDING AQUATIC ORGANISMS

Study	Objective	Algorithms	Performance Results
[27]	Improve feeding accuracy using ARIMA and IoT	ARIMA	±0.34g feeding accuracy, 123.12% growth rate
[28]	Optimize feeding using fuzzy logic	Fuzzy Logic	Improved fish health, reduced mortality
[29]	Integrate computer vision with FNN	YOLOv5 + FNN	80-90% accuracy in feeding behavior analysis

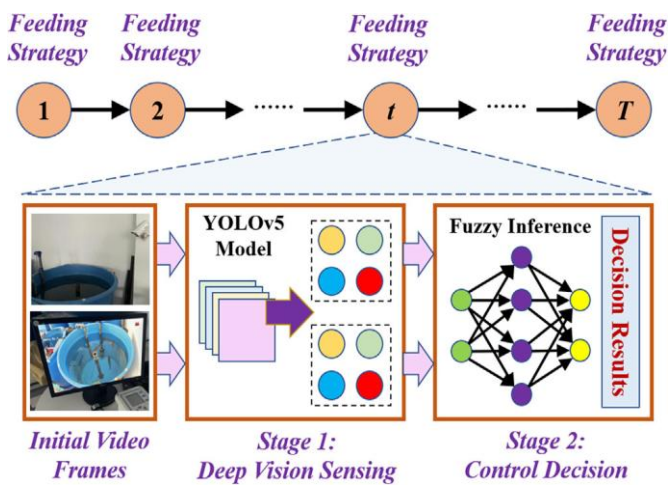

Fig. 5. Intelligent feeding control framework using deep learning and fuzzy logic for [29] work

Figure 5 illustrates the decision-making stages in a system [29]. In the first stage, deep vision sensing is performed. The inputs are characterized as videos, which are divided into frames. The YOLOv5 model is then used to analyze each frame to identify objects and important activities. The second stage involves a fuzzy inference system using YOLOv5 results and fuzzy rules to output automated control decisions in the system.

Feeding systems lack comprehensiveness and diversity of information sources. Relying on a single source, such as images or weight sensors, is not sufficient to accurately predict the required quantity and timing. Different sources must be integrated, and a system must be built that adapts over time, as parameters and conditions are subject to change, such as the number of fish, fish size, tank content, and weather.

III. DISCUSSION

Previous research on aquaponics and aquaculture systems offers advanced solutions using smart and advanced technologies to improve production efficiency, resource consumption, and reduce human errors in several areas, including in the field of intelligent control, which work in this area shows the importance of maintaining environmental variables in a stable degree, such as improving water quality and ensuring nutrient quality, with the support of technology and cloud analytics to enable real-time monitoring and minimize human intervention. Decision support systems using models such as digital twins, neural networks, and utilizing a previous data set to achieve a high ability to identify positive decisions by analyzing data and predicting environmental conditions to ensure the efficiency and safety of the system. Smart feeding systems ensure the health of organisms and reduce the risk of overfeeding. Techniques such as computer vision and predictive models are used to determine feeding times and minimize the waste of resources. The new optimization technique focused on several areas, including managing resource consumption and improving productivity such as plant growth rate and crop quality. Predictive modeling in intensive system management also played an important role in predicting water quality and growth and fighting harmful algae to ensure stable environmental conditions. In Table 15 you can see the sensors used in each area of smart systems in aquaculture and aquaponics. However, these systems lack a number of elements, including the comprehensiveness of the criteria and sources used and the failure to address the most important biological criteria, weather criteria, and even lighting and solid elements. The work [23] also presented 21 variables in its study, demonstrating the strength of its results. In addition, few systems rely on measuring nitrate and ammonia, despite their importance in the system, which weakens the control systems and decision-making. In addition, the research relied on specific organisms, which reduces generalisability, and few systems included hybrid algorithms such as fuzzy logic, machine learning, and deep learning, which can solve the problem of multiple conditions and different data sources. In addition, no mention was made of achieving a balance between the needs of plants and aquatic organisms in aquaponics systems.

TABLE. 15 THE TABLE SHOWS THE STANDARDS USED IN EACH AREA OF SMART AQUACULTURE AND AQUAPONICS SYSTEMS.

Sensor	Smart Control	IoT Monitoring & Control	Decision Support	Smart Feeding	Techniques for improving	Predictive Modeling
pH Sensor	✓	✓	✓	✓	✓	✓
Temperature Sensor	✓	✓	✓	✓	✓	✓
Turbidity Sensor	✓	✓	✓	✗	✗	✓
EC Sensor	✓	✓	✗	✗	✗	✗
TDS Sensor	✗	✓	✗	✗	✗	✗
DO Sensor	✓	✓	✓	✗	✓	✓
Salinity Sensor	✗	✓	✓	✗	✓	✗
Humidity Sensor	✗	✓	✗	✗	✗	✗
CO_2 Sensor	✗	✓	✗	✗	✓	✗
Ammonia Sensor	✗	✓	✗	✗	✓	✗
Water Flow Sensor	✗	✗	✓	✗	✗	✗
Light Sensor	✗	✗	✓	✗	✗	✗
Load Cell	✗	✗	✗	✓	✗	✗
RTC	✗	✓	✗	✓	✗	✓
Ultrasonic Sensor	✗	✗	✗	✓	✗	✗
Vision Sensors (YOLOv5)	✗	✗	✓	✓	✗	✓
Water Hardness Ca^{2+} Mg^{2+}	✓	✗	✗	✗	✗	✗
Nitrite Sensor	✓	✗	✗	✗	✗	✗
Ultrasonic sensor (HC-SR04)	✗	✗	✗	✓	✗	✗

IV. CONCLUSION

Aquaponics and aquaculture systems are emerging as strategic solutions for achieving food security and improving production quality. At the local level, local aquaculture projects have emerged, such as the Sarl Titri Aquacole farm [30] in Benchicao (Médéa), a freshwater fish production project, and the Litopenaeus vannamei shrimp farming project[31] in Hassi Ben Abdallah (Ouargla), which is a pilot station using the Biofloc (BFT) system with low-salinity groundwater, demonstrating the importance of aquaculture in the food economy. In addition, there are proposed projects such as Saharaponics[32] to support desert agriculture with the Aquaponics system. However, these projects lack support for modern technologies, which have provided multiple solutions, including environmental stability through monitoring systems, as well as automated control systems that have contributed to reducing human error and adapting to environmental changes. These systems have also supported system improvement methods, including nutrient optimization, resource control, and crop growth monitoring, as well as systems that support farmers' decisions that can adapt to their needs by analyzing system failures. Nutrition solutions for aquaculture systems have also been proposed, aimed at conserving resources, with these ideas supported by environmental forecasting technologies that enable proactive risk prediction. However, there is a noticeable missing element, which is that these works lack comprehensive criteria for contribution, decision-making, and smart control, such as not taking into account the importance of ammonia, nitrates, and solids in hydroponic systems, given their importance in the life of aquatic organisms. These substances can cause toxicity in hydroponic water, leading to system failure. Furthermore, systems that focus on nutrient recovery are few, which affects the sustainability of the system and its need for other fertilizers. Therefore, this topic remains under study to determine whether there is a need to find alternative solutions that are more efficient, more comprehensive in environmental standards, more accurate in results, and more adaptable.

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